Wengqiang Wu

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EDUCATION

Southern University of Science and Technology (SUSTech) 08/2023-06/2027

- B.S. in **Computer Science and Technology**
- **GPA:** 3.78/4.0 | **Average Score:** 90.18/100 | **TOEFL:** 110/120 | **GRE**: 328 (V: 160, Q: 168)
- **Honors:** Second-Class Scholarship, 2023

University of California, San Diego (UCSD) 03/2026-06/2026

- Non-degree Exchange Program | **GPA:** 3.67/4.0
- **Coursework:** AI: Probabilistic Models; Machine Learning for Music; Intro to Autonomous Vehicles

PUBLICATION

[1] Li, J., Li, Z., **Wu, W.,** Chang, Z., Gao, M., Yang, J., & Zheng, F. **DeSeG: Decoupling Semantic Intent and Geometric Constraints for Physically Plausible Human-Scene Interaction.** *WACV 2027.* (Accepted)

[2] Cao, H., **Wu, W.**, Tu, K.-T., Otnes, M., Delmerico, J., Wang, R., Tian, Y., & Atanasov, N. **HIGS: Hierarchical Implicit Grids for Joint Geometric and Semantic Scene Understanding.** *Submitted to IJRR, 2026.*

RESEARCH EXPERIENCE

Existential Robotics Laboratory, UCSD 03/2026-Present

Undergraduate Researcher, Advisor: Prof. Nikolay A. Atanasov

Vision-Language Mapping & 3D Grounding | *Core Member, Submitted to IJRR, 2026.*

- Developed a **hashgrid-based framework** for projecting 2D Vision-Language Model (VLM) features into 3D space to support real-time and trainable online semantic mapping.
- Integrated a pre-trained Locate-3D DETR-style decoder for text-prompted 3D semantic grounding.
- Designed a hybrid semantic-geometric pipeline incorporating Signed Distance Field (SDF) and VLM features to improve 3D grounding, pose estimation, and submap alignment.
- Project page: https://existentialrobotics.org/HIGS_webpage/

SUSTech Visual Intelligence Laboratory 03/2025-Present

Undergraduate Researcher, Advisor: Prof. Feng Zheng

Depth-Based Locomotion Policies for Complex Terrains | *Core Member, Manuscript in Prep.* 06/2026-Present

- Trained a **depth-based footstep-tracking policy** for robot locomotion to traverse complex environments, including gaps, stairs, and stepping stones.
- Embedded target footholds into depth maps, allowing the policy to jointly track target footsteps and adapt its pose and subsequent goal commands from visual depth.
- Completed sim-to-real transfer and validated the policy on physical robotic hardware.

Hierarchical Vision-Language-Action framework | ***Co-First Author, Accepted to WACV*** 12/2025-03/2026

- Independently Implemented a **Residual CVAE** and designed a **Geometry-Aware Cross-Attention** module to fuse CLIP text/goal representations with 3D scene voxels to form a decoupled semantic prior.
- Developed a residual sampling scheme that models kinematic residuals over the semantic prior using a customized KL-divergence objective.

- Adapted a two-stage generative pipeline that conditions a physics-regularized DDPM on VL-prior latents for realistic and collision-free robot interaction.

Robot Pose Retargeting from Multi-source Human Data to Simulation | *Core Member* 03/2025-12/2025

- Applied GMR algorithm to retarget SMPL-formatted human motion from LAFAN1 and AMASS to the Unitree H1 humanoid robot.
- Trained physics-based motion-tracking policies in Isaac Lab using the BeyondMimic framework.
- Generated and extracted additional human-motion data using MDM from text descriptions and GVHMR from raw videos.

INTERNSHIP EXPERIENCE

Spatialtemporal AI, China 07/2025-09/2025

Development Engineer, Mentor: Kaidong Zhang

- Developed a **vision-language-guided robotic grasping system** by integrating Grounding DINO with an LLM, allowing robotic arms to execute grasping tasks from natural-language instructions.
- Developed and optimized a bimanual robotic teleoperation system using Apple Vision Pro to support real-time control of physical robotic arms and synchronized collection of demonstration data.
- Integrated YOLO-based chess-piece recognition with a chess engine to develop an autonomous Xiangqi-playing robotic system, enabling the robotic arm to perceive board states and interact with human opponents.

SELECTED PROJECTS

Blockchain Simulation with Sockets 05/2025-06/2025

Core Member, Advisor: Prof. Zhuozhao Li

- Implemented message routing, blockchain verification/storage, and dynamic node discovery in a socket-based distributed system.
- Developed RTT measurement and link-state routing for network path optimization.

Single-Cycle RISC-V CPU in Verilog 05/2025-06/2025

Core Member, Advisor: Prof. Yuhui Bai

- Designed and integrated core CPU components, including instruction fetch, controller, decoder, and memory I/O modules.
- Implemented UART communication logic and conducted functional validation of the complete processor.

EXTRACURRICULAR ACTIVITIES

Member, SUSTech Drawing Club 11/2023-11/2025

- Participated in art exchange events and public exhibitions.

Volunteer, SUSTech Annual Concert 11/2023

- Coordinated rehearsal schedules with instrumental groups and provided logistical support.
- Assisted with rehearsal guidance, stage management, and on-site event operations.

TECHNICAL SKILLS

- **Programming:** Python, C++, Java, Rust, Verilog
- **AI/ML:** PyTorch, Computer Vision, Vision-Language Models, LLMs, Diffusion Models
- **Web Development:** JavaScript, CSS, HTML
- **Database:** MySQL, PostgreSQL